

THE DISPLAY OF THE MOTHER BOARD

INTRODUCTION

The purpose of this report is to provide an overview of the display components associated with the motherboard we carried out in the lab. The motherboard is the most important component of a computer system that Connect various hardware devices and facilitate communication between them. while the motherboard itself does not possess a display , it incorporates several interface and connector to support the connection of display devices. This is the display of motherboard form factor.



- We carryout our PC HARDWARE ASSEMBLY practical on the 27/01/2024,

we were divided in to groups of fives and we where ask to display the motherboard and take pictures of each of the part we remove after which we will have to reassemble it back.

After we had display the parts of the motherboard our supervisor came and ask us some few questions about the components of the motherboard . This our the questions we where evualated on.

- 1) He asked us to identify the parallel port,RJ45,USB,HDMI,DVI,VGA and anolog Audio.

- 2) He also ask us to identify the processor, the Northbridge, Southbridge, the main board battery(CMOS battery), the fan, heat sink, RAM, memory slots
- 3) we where ask to give the differences between the CMOS and the CMOS battery.
- After we answered the question we started assembling back the part of the motherboard , when we where done the supervisor came and checked if we hand assemble it well. After which he graded our marks.

- 1) During our practical we face problems when we were asked to give the differences between the CMOS and CMOS battery. The differences we gave the supervisor said there were not correct.
- 2) we face difficulty to identify a parallel ports.

Here are the differences between CMOS and the CMOS battery

1) The CMOS is responsible for storing essential system information such as date and time and also BIOS settings while CMOS battery it provides power to a special type of memory called CMOS memory

HERE ARE THE TYPES OF INPUT AND OUTPUT PORTS WE SAW ON THE MOTHERBOARD



- a) **Parallel port:** the primary function is to facilitate the transfer of data between a computer external devices such as printer, scanners. Parallel port transfer multiple bits of data simultaneously.
- b) **DVI port:** it is use to connect to digital video interface display device.

- **c) VGA port:** it is use to connect a video graphic array display devices
- **d) RJ45 port:** it is use to connect ethernet network.
- **e) USB :** it is use to connect external devices with universal serial bus connector such as mice, printer.
- **f) HDMI:** it is use to connect high definition multimedia interface.
- **g) Audio:** it is use to connect to a speaker system.
- **h) Ps/2 ports:** there are use to connect keyboard and mice.the ps/2 ports provide a reliable and standardize connection for input devices.

THE MOTHERBOARD AND IT'S COMPONENTS

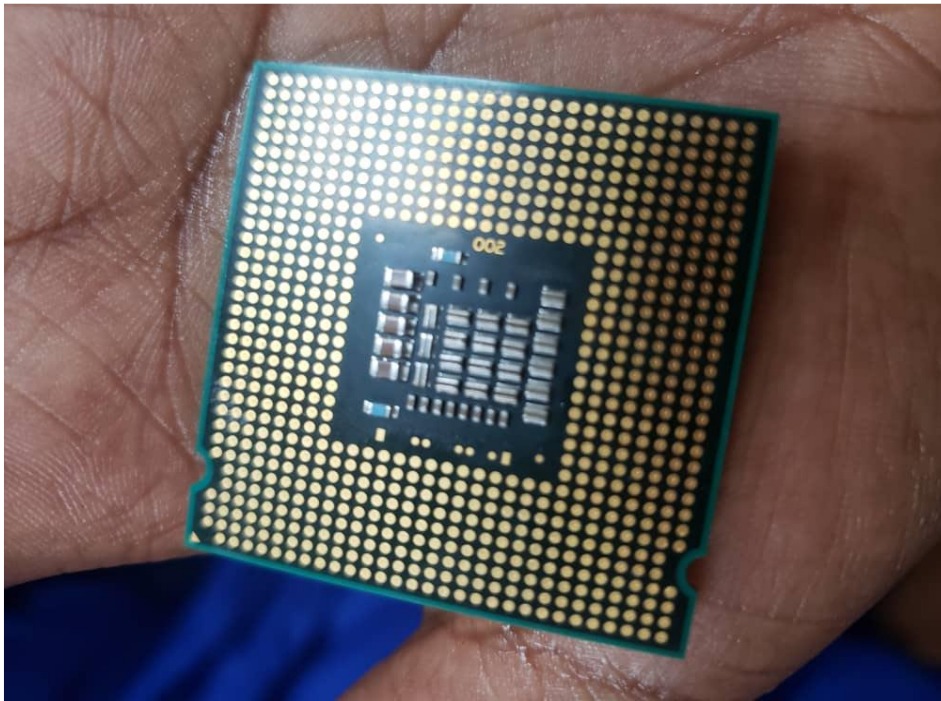
1) **The hard disks drive:** the hard disk drive is the hardware component that stores and retrieve all of your digital content or data. the hard disk dive provide long term, non volitile storage for various type or data including the operating system, software application.



- The hard disk drive



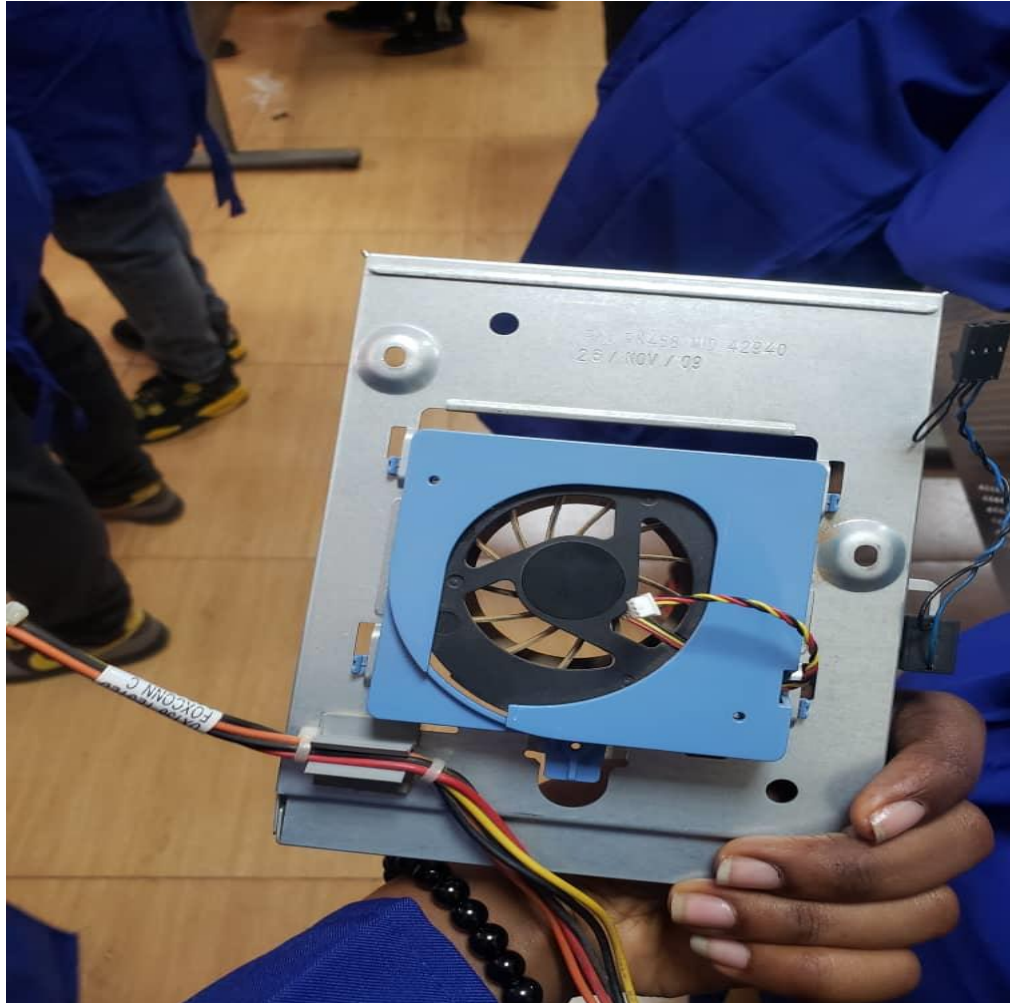
- **processor:** a central processing unit is the electronic circuitry that execute instructions comprising a computer program.the cpu preform basic arithmetics, logic,controlling and input/output operation specify by the instruction in the program.



The processor

- **Fans:** each motherboard has several connector for fans, especially the fans and the case fan. the CPU fan are specially designed to cool down the processor. There are typically mounted directly on top of the CPU, dissipating heat generated by the processor's operation. The case fan maybe placed as intake fans, drawing cooler outside air in through the front or bottom of the chassis.

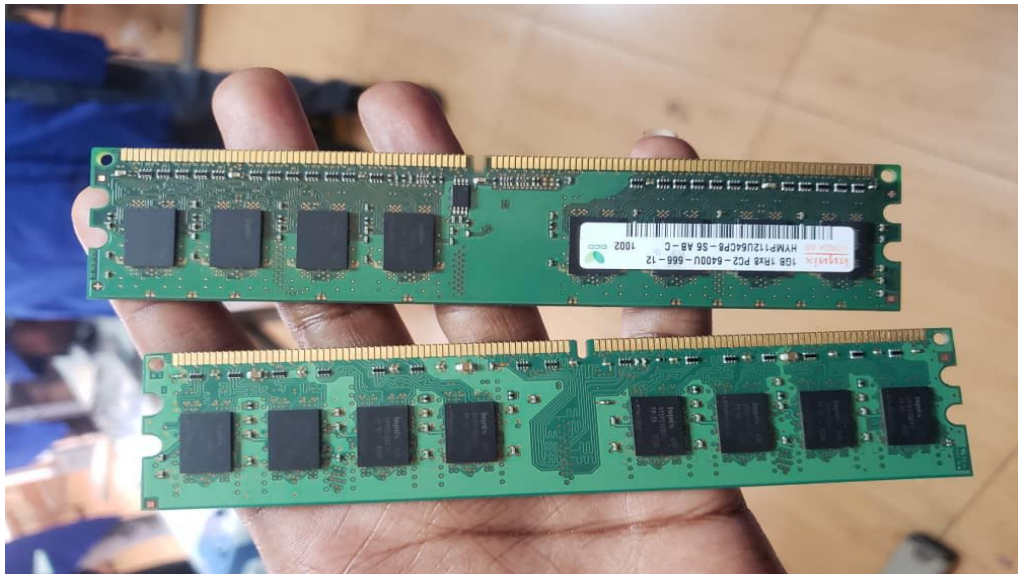
The CPU fan and the case fan



- **Heat sink:** a heat sink is an essential component in electronic devices and systems that are prone to generate heat during their operation. The function of the heat sink is to dissipate or remove excess heat generated by the electronic component such as processor, graphics card or power amplifiers , and maintain temperature within an acceptable range.



- **RAM:** the random access memory , is a crucial component in a computer system that plays a vital role in its overall performance. RAM is a type of volatile memory that provide s temporal storage for data and instruction that are actively being used by the computer's processor. The RAM allows the computer to perform most of it everyday tasks, such as loading, applications and editing a spreadsheet.



THE RAM

The layout of the motherboard



- 1) **CPU socket:** the CPU socket is where the central processing unit (the brains of the computer) is inserted. The CPU socket is a physical component on a computer's motherboard that serves as a connection point for the central processing unit. The main function of the CPU socket is to provide both physical and electrical connections between the CPU and the motherboard

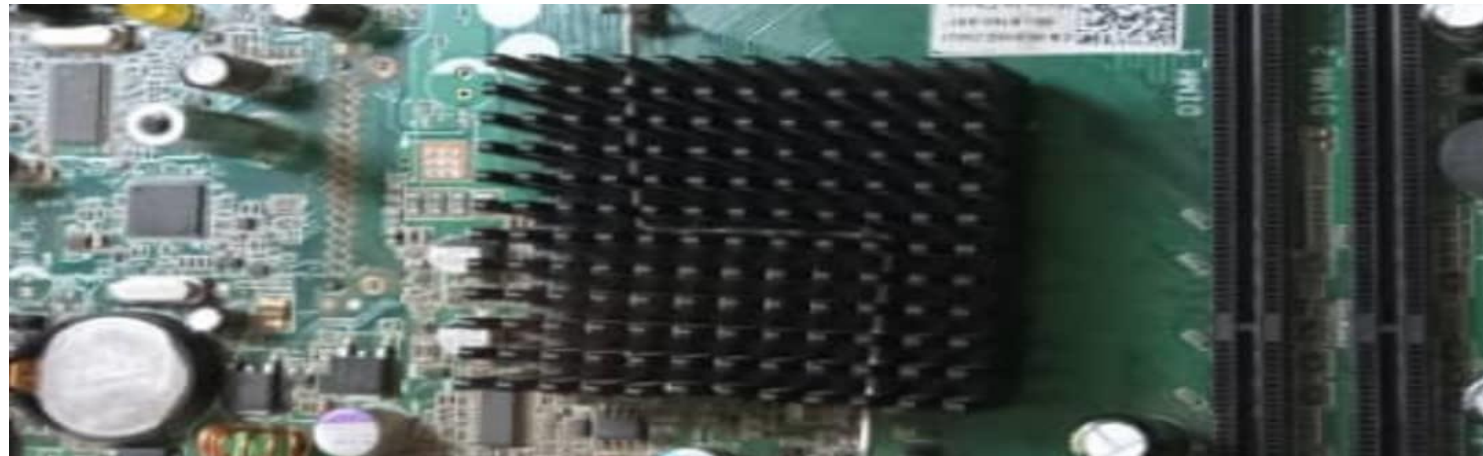


- **2)Memory slots:** These slots are where extra Random access memory(RAM) is inserted to provide capacity to the computer. The memory slots provide a mechanical interface for installing and securing memory module onto the motherboard. The memory slots contains a set of electrical contacts or pins that establish electrical connection between the memory module and the motherboard



- **3)North Bridge** : The northbridge is the controller that interconnect the CPU to memory via the frontside bus(FSB) it also connects peripherals via high-speed channels such as PCI Express, The northbridge may include a display controller, obviating the need for a separate graphic cards.

- Northbridge



- **4)Southbridge:** the southbridge is an intel chipset that manages the basic forms of input/output (I/O) such as universal serial bus (USB) ,serial, audio, integrated drive electronics (IDE) industrial standards architecture (ISA) input/output in a computer.
- The southbridge serve as connector between the CPU and lower-speed components of the computer.



- **5)The main board battery:** the main board battery also known as the motherboard battery or the CMOS battery serv as a crucial function in a computer. The CMOS battery is responsible for providing power to the complementary metal-oxide semi conductor (CMOS) chip on the mother board, the CMOS batter powers the real time clocks which keeps track of current date and time.



- **6)24 pin main power connector:** The 24 pin main power connector is what powers the entire motherboard. The 24 pin main connector delivers power to the motherboard, supplying the necessary voltage and current for its proper operation.
- **7)SATA connector:** A SATA connector is a standardized interface used to connect storage devices ,such as hard disk drive (HDD), solid state drive (SSD),and optical drive to a computers motherboard.



conclusion

The motherboard is an essential component in assembling a pc hardware system.

The practicals on pc hardware assembly has provided valuable hand on experiment and knowledge in building a computer system. Through the practicals session, we learned to follow a proper assembly procedures.

The practicals also help us to gain more understanding of the various hardware component such as the motherboard, processor, memory , storage devices and power supply .

The pc hardware assembly practicals was a success, even though we had some few challenges.

Thanks to the pc hardware assembly lecturer who stood by us thanks sir

